VOLODYMYR POLISHCHUK

Head of Software & Autonomy | Systems Architect | GPS-Denied UAV Navigation

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SUMMARY

Systems architect and tech lead with 19 years in production engineering: Wix's serverless developer platform, used by about 3,000 developers across 2,000+ microservices; a real-time betting engine taking 1M requests/s at peak at Playtech; and payment-network infrastructure at UkrCard. Since 2026 I have headed software and autonomy at Applied Technology Collective, a Ukrainian UAV developer. I joined as its only software engineer, built GPS-denied navigation, now in production, that runs on a $15 companion computer, with its own flight-controller firmware and simulator, and hired the team. I start from the physics and the measurements, not from frameworks. I build AI-native: I own the architecture, the estimation math and the acceptance tests, and direct AI coding agents as the implementation team.

EXPERIENCE

Applied Technology Collective | Head of Software & Autonomy — 2026 – Present

Ukrainian developer of strike, loitering-munition and cargo UAVs. I joined as the only software engineer, built the software and autonomy function, and hired two engineers. NADIR is in production, SEXTANT is rolling out, and ORRERY is the standalone simulator both are tested in. All three are my own IP, which the company uses. Code walkthrough on request.

NADIR — position hold for ArduPilot from one downward camera

- A Rust daemon on a $15 companion computer turns one downward camera, with the autopilot's IMU, into the drone's velocity and a position anchored to the ground below, which ArduPilot's EKF fuses in place of GPS. Shipped as a product, not a demo: a sealed SD image, a web console, calibration from QGroundControl, and pre-arm checks that refuse flight until the setup is right.
- Approach: every instability is traced to its physical cause rather than tuned away; the daemon moved from C to Rust one module at a time, with the drone flyable after every step; and its own ArduPilot firmware build turns on the vision fusion that stock builds for small flight controllers leave out.

SEXTANT — long-range navigation by terrain geometry

- Absolute position and heading without GPS, from the shape of the terrain rather than its appearance: the forward camera's skyline is matched against a terrain model and downward motion stereo measures the relief below, both fused by ArduPilot's EKF. I chose geometry after prototyping and retiring satellite-image matching (SILO): appearance changes with snow, foliage and sun, terrain geometry does not.

ORRERY — standalone simulator for camera-based flight

- A Godot flight simulator in which NADIR and SEXTANT fly their acceptance tests. It runs any ArduPilot SITL build and models cameras as real sensors; the SITL, every camera and the vehicle are set entirely by configuration.

Wix.com | Principal Software Engineer / Architect — 2017 – 2026

- Led Wix's adoption of gRPC and Protobuf as the contract for its internal APIs, the foundation for opening them to third-party developers and the Wix App Market; built the Protobuf linter in the core CI/CD pipeline that every service's API passes, with code and documentation generated from the contract.
- Designed and taught the asynchronous-programming courses and workshops, mandatory for Wix's 1,000+ server engineers, that moved the backend to async code and cut service resource use.
- Interviewed weekly throughout my time at Wix (300+ interviews), at all three stages of the company-wide hiring loop: system design, coding, and the Server Guild deep-dive into a candidate's experience and interests that decides where in the company they fit. Sat on the hiring committees: scored and voted on each candidate and helped shape the new hire's career path.

- Architected the internal serverless platform (App Engine) and built it with a core team of five. It let about 3,000 developers, mostly front-end, ship Node.js services across 2,000+ microservices without touching Kubernetes, service wiring or CI/CD; control plane in Scala and ZIO with typed errors.
- Designed a throughput-based autoscaler beyond what KEDA/HPA offer: it pre-warms capacity before datacenter switches and adjusts CPU and memory live, scaling without downtime.
- Architected the Wix App Market's AI chat agent: retrieval-augmented generation on Vespa, and a bidirectional streaming pipeline from WebSockets to a gRPC server for real-time LLM responses.
- Domains: automated SSL certificate issuance and renewal for 8M+ domains with no manual steps; architected Wix's ICANN-accredited registrar platform and an in-house DNS server; led the migration of the legacy Java domain-management monolith to microservices.

Playtech | **Software Architect** / **Tech Lead** 2012 – 2017

- Tech lead of a team of five engineers (a team leader handled management): I set the design, the technical direction and the goals, and ran the team's internal communication. Led the sports-betting platform's migration from a Java EE monolith to a reactive Scala architecture (Scalaz, algebraic domain modeling).
- Real-time streaming held 60M requests/min peaks during the FIFA World Cup over WebSockets and raw sockets, with back-pressured Akka Streams keeping memory bounded under load.
- Cut the bare-metal footprint to a highly available 3-node minimum, with New Relic observability.
- Regulation ruled out the cloud, so I designed a Dynamo-style, eventually consistent key-value store on LevelDB over bare-metal disks, with Akka Cluster gossip for membership and state machines for replica logic; reads are served locally, with no network round-trip.

UkrCard | **Systems Architect** 2007 – 2012

- Led the move of every ATM, POS and self-service terminal on the payment network from wired X.25 to wireless TCP/IP, with end-to-end VPN and SSL encryption (2008–2012).
- Fraud-detection assistant (Master's project, 2007–2008): a neural network written from scratch in Java, ensembled with a human-in-the-loop meta-model that learned from operator feedback to cut false positives under severe class imbalance.

SKILLS

Autonomy: visual and visual-inertial odometry, EKF sensor fusion, camera calibration and timing, terrain-referenced navigation, ArduPilot (EKF3, firmware, SITL), MAVLink.
Languages: Rust, C, Scala (ZIO, Akka, Scalaz), Java, Python, Node.js.
Systems: embedded Linux on ARM, real-time pipelines, Kubernetes, gRPC/Protobuf, distributed storage, CI/CD, Godot.
AI: RAG, LLM applications, neural networks from first principles.

EDUCATION

M.Sc. in Cybernetics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics. Thesis: Ensuring the Dependability of Transaction Classification in a Payment System.

B.Sc. in Applied Mathematics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics

MILITARY & OTHER QUALIFICATIONS

Armed Forces of Ukraine (reserve): Junior Lieutenant, military specialty 708200: operation, tactical deployment and maintenance of air-defense radar systems.
IYT Skipper (International Yacht Training): practical navigation.
Interests: world models for robot navigation (V-JEPA, Genesis); basal cognition (Michael Levin); foundations of physics.